a

Congenital phenotypes

HPO phenotype

GM2−ganglioside accumulation
Histiocytosis
Short telomere length
Spinocerebellar atrophy
Neonatal sepsis
Profound static encephalopathy
Reduced beta−hexosaminidase activity
Subependymal giant−cell astrocytoma
Abnormality of amino acid metabolism
Decreased activity of mitochondrial complex I
HbS hemoglobin
Decreased CSF biopterin level
Decreased mitochondrial number
Profound global developmental delay
Increased circulating very long−chain fatty acid concentration
Loss of Purkinje cells in the cerebellar vermis
Absence of bactericidal oxidative respiratory burst in phagocytes
Defective production of NFKB1−dependent cytokines
Progressive spastic paraparesis
Recurrent Aspergillus infections

profound
severe

b

e

Top targets

GM2−gangliosidosis, AB variant	alternatively activated macrophage
Immunodeficiency 27A, mycobacteriosis, AR	microglial cell
Dyskeratosis congenita, autosomal recessive, 3	T cell
Spinocerebellar ataxia 1	alternatively activated macrophage
3−Methylglutaconic aciduria with deafness, encephalopathy, and...	chondrocyte
Hypotonia, infantile, with psychomotor retardation and charact...	fibroblast
Sandhoff disease	extravillous trophoblast
Tuberous sclerosis−2	endocrine cell
Tyrosinemia type 2	epithelial cell
Mitochondrial complex I deficiency, nuclear type 12	inflammatory cell
Hereditary persistence of fetal hemoglobin−sickle cell disease...	erythroid lineage cell
Alacrimia−choreoathetosis−liver dysfunction syndrome	mature neutrophil
DNA2−related mitochondrial DNA deletion syndrome	enterocyte
Mental retardation, x−linked syndromic, Christianson type	GABAergic neuron
Mental retardation, enteropathy, deafness, peripheral neuropat...	gastric goblet cell
Spastic ataxia, Charlevoix−Saguenay type	neuron
Granulomatous Disease, Chronic, Autosomal Recessive, Cytochrom...	monocyte
Ectodermal dysplasia and immunodeficiency 2	stromal cell
Adrenomyeloneuropathy	stratified epithelial cell
Granulomatous Disease, Chronic, Autosomal Recessive, Cytochrom...	monocyte

f

Orthologous genes

Homo sapiens	Macaca mulatta	Mus musculus	Danio rerio	Drosophila melanogaster	Caenorhabditis elegans
GM2A	LOC713375	Gm2a	gm2a		
IFNGR1	IFNGR1	Ifngr1			
WRAP53	WRAP53	Wrap53	wrap53	WDR79	
ATXN1		Atxn1	atxn1a		
SERAC1	SERAC1	Serac1	serac1	CG10383	
UNC80	UNC80	Unc80	unc80	CG18437	
HEXB	HEXB	Hexb	hexa		hex−1
TSC2	TSC2	Tsc2	tsc2	gig	
TAT	TAT	Tat	tat	CG1461	tatn−1
NDUFA1		Ndufa1	ndufa1		
HBG1					
NGLY1	NGLY1	Ngly1	ngly1	Pngl	png−1
DNA2	DNA2	Dna2	dna2	CG2990	dna−2
SLC9A6	SLC9A6	Slc9a6	slc9a6a	Nhe3	nhx−5
AP1S1		Ap1s1	zgc:162858		
SACS	SACS		sacs		
CYBA	LOC696748	Cyba	cyba		
NFKBIA	NFKBIA	Nfkbia		cact	
ABCD1	ABCD1	Abcd1	abcd1		
CYBA	LOC696748	Cyba	cyba		

c

Non−congenital phenotypes

Occurrence
never
rarely
often
always

Severity score
33
30
27
24

Cerebellar medulloblastoma
Vegetative state
Malnutrition
Hypoglycemic encephalopathy
Decerebrate rigidity
Acute myeloid leukemia
Pallidal degeneration
Cerebral cortical neurodegeneration
Neuronal loss in central nervous system
Cerebral vasculitis
Acute myelomonocytic leukemia
B Acute Lymphoblastic Leukemia
Late−onset spinocerebellar degeneration
Increased CSF protein concentration
Progressive leukoencephalopathy
Progressive psychomotor deterioration
Esophageal carcinoma
Glucocortocoid−insensitive primary hyperaldosteronism
Anorexia
Kernicterus
Necrotizing myopathy

death
intellectual disability
impaired mobility
blindness
physical malformations
sensory impairments
immunodeficiency
cancer
reduced fertility
congenital onset

profound
severe

d

g

Disease	Cell type
Familial multinodular goiter	neuron
Mitochondrial DNA depletion syndrome 4A (Alpers type)	kidney loop of Henle epithelial cell
Visceral myopathy 1	smooth muscle cell
3−Hydroxyacyl−Coa dehydrogenase deficiency	inflammatory cell
Krabbe disease	alternatively activated macrophage
Platelet disorder, familial, with associated myeloid malignancy	mature neutrophil
Hypoprebetalipoproteinemia, acanthocytosis, retinitis pigmento...	primordial germ cell
Mitochondrial DNA depletion syndrome 4A (Alpers type)	kidney loop of Henle epithelial cell
Huntington disease	alternatively activated macrophage
Hyper−IgE syndrome 2, autosomal recessive, with recurrent infe...	myeloid cell
Ataxia−Pancytopenia syndrome	alternatively activated macrophage
Thrombocytopenia 5	hematopoietic stem cell
Hemolytic anemia due to gamma−glutamylcysteine synthetase defi...	erythroid progenitor cell
Creutzfeldt−Jakob disease	astrocyte
Leukoencephalopathy, progressive, with ovarian failure	visceromotor neuron
GM1−gangliosidosis, type II	extravillous trophoblast
Tylosis with esophageal cancer	alternatively activated macrophage
Generalized pseudohypoaldosteronism type 1	taste receptor cell
Argininemia	mature neutrophil
Crigler−Najjar syndrome, type I	enterocyte
Hereditary myopathy with early respiratory failure	cell of skeletal muscle

h

Homo sapiens	Macaca mulatta	Mus musculus	Danio rerio	Drosophila melanogaster	Caenorhabditis elegans
KEAP1		Keap1	keap1b	Keap1	
POLG	POLG	Polg	polg	tam	
ACTG2	ACTG2	Actg2	acta2	Act79B	
HADH	HADH	Hadh	hadh		
GALC	GALC	Galc	galcb		CELE_C29E4.10
RUNX1	RUNX1	Runx1	runx1		
PANK2	LOC717639	Pank2	pank2		
POLG	POLG	Polg	polg	tam	
HTT		Htt	htt		
DOCK8	LOC698081	Dock8	dock8		
SAMD9L	LOC701572	Samd9l			
ETV6	ETV6	Etv6	etv6		
GCLC	GCLC	Gclc	gclc	Gclc	gcs−1
PRNP	PRNP	Prnp			
AARS2	AARS2	Aars2	aars2		
GLB1	LOC709355	Glb1	glb1		
RHBDF2	RHBDF2	Rhbdf2			
SCNN1A	SCNN1A	Scnn1a			
ARG1	ARG1	Arg1			
UGT1A1	UGT1A1	Ugt1a1		CG10178	
TTN	LOC703527	Ttn	ttna		